BUS 520

Global Markets, Investments & Analytics | Course Weeks 7-9

Learning Objectives

By the end of these three weeks, you should be able to:

1. Describe data and explain what they can and cannot tell us.
2. Use financial and customer measures to explain business results.
3. Run prepared models and explain their results.
4. Use AI to write, run and check code (vibe coding).

Our main question

How can we use data and AI to make a better business decision?

Our path: understand the evidence -> use it for a business question -> compare better ways to answer that question.

AI is a **decision co-worker**. It can help explain code, make a first draft and fix errors. People still choose the question, check the evidence and take responsibility for the final decision.

You do not need **coding experience**. Our goal is to understand, check and use analysis, not to become software developers.

The three-week path

Week	Main question	What you will learn to do
Week 7: Understand data	What do the data show, and how much can we trust the result?	Describe data, learn from samples, run a first model and check a dashboard.
Week 8: Use data for business	What is happening to performance, customers and demand?	Read returns and risk, use business measures, study customers and make a first forecast.
Week 9: Modern machine learning	Which machine learning method fits the problem, and how well does it work on new data?	Compare models, find groups, make data simpler and explore text and AI methods.

What we will cover

Week 1: Business Analytics and Tools

Question: How do we turn data into useful evidence?

- **BUS 520 and AI:** our learning plan, AI as a co-worker, and our responsibility to check results.
- **Theory and practice:** use models to make reality simpler, use PCA to see what we keep or lose, and solve the right business problem.
- **Tool choice:** Python, R, Julia, Power BI and Excel. Their uses overlap; choose a tool for the task.
- **Digital economy:** DX and AX, the gold ETF example, and numbers, text, images and sound as data.
- **Understanding data:** mean, deviation, variance and standard deviation. Next, population, sample, standard error, the Central Limit Theorem (CLT) and a simple confidence interval.
- **First models:** regression predicts a number; classification predicts a category. Keep test data separate, compare a simple baseline and explain errors.

Practice: run the prepared Colab notebook, change a marked setting, export a CSV, check a local dashboard and explain a result.

Week 2: Performance Analysis and Customer Analytics

Question: How can analysis help a manager understand results and plan action?

- **Financial performance:** returns, risk and dashboard reading. A high return alone does not show all the risks.
- **Sales and customers:** key performance indicators (KPIs), customer groups and decision dashboards. Give each measure a clear meaning and unit.
- **Customer choice:** use classification to study a choice, such as buying or not buying. A prediction is not an observed purchase.
- **Sales and demand:** use time-series data to forecast later demand. Train on earlier data and test on later data.

Planned practice: explain a business measure, compare customer groups and read a prepared classification or forecast result. State what the result can and cannot support.

Week 3: Machine Learning Analytics

Question: Which model gives useful results on new data?

- **Supervised learning:** regression, random forests and boosting. These models learn from examples with known answers. Compare test results with a simple starting model.
- **Unsupervised learning:** clustering and PCA. Find groups or use fewer dimensions when there are no known answer labels. Explain what the groups or simpler data mean.
- **Deep learning:** ANN, RNN, LSTM and Transformers. Learn their basic uses through prepared examples. You will not build every model from the beginning.
- **Text analytics:** natural language processing (NLP) and sentiment analysis. Explore how tools work with text and estimate its tone. Check examples where a tool gets the meaning wrong.

The workflow we will use each week

1. **Open the course files on GitHub or Moodle.**
2. **Work in Google Colab.**
3. **Export a CSV and check its contents.**
4. **Ask Gemini or another available AI tool for code.**
5. **Run Streamlit on your laptop.**
6. **Check the dashboard values against the CSV.**
7. **Explain the business meaning of the results.**

All analysis, statistics and model fitting stay in Colab. Only the dashboard runs locally. You do not need local Jupyter or an IDE.

- If laptop setup is slow or you cannot run the app, continue the Colab activity.
- Ask the teaching assistants (TAs) for help.

How we will learn

Run -> Change -> Observe -> Interpret

- Start by running prepared code that works.
- Change one setting, not the whole program.
- Observe which numbers or charts change.
- Explain what the change means in simple business language.
- Work in your assigned team and take turns. Each student should explain at least one measure.
- Use AI for coding support. Your team must write its own interpretation of the results.

What good work looks like

You should be able to answer five simple questions:

1. **Identify the business problem:** What problem are we trying to solve?
2. **Examine the evidence:** What data and measures help us answer the question?
3. **Evaluate the result:** What metrics and checks help us judge the quality of the result?
4. **Interpret the result:** What does the result mean, and what are its limits?
5. **Plan implementation:** What should the manager do next, and how will we check whether the action works?

We will look for clear questions, suitable data and measures, careful checks, and useful explanations. A dashboard shows evidence. The manager uses that evidence to make a decision.

Practice and assessment

- **Teamwork:** The professor will assign students to teams. Complete all in-class assignments and practice activities in these teams.
- **Group project:** At the end of the course, each team will complete a project that connects Business Analytics with the two earlier sections: Global Markets and Finance.
- **Assessment:** Weekly team practice will help you prepare for the group project and final exam. It adds no separate assessment weight. The official course weights remain **group project 40%** and **closed-book final exam 60%**.